Drug–Receptor Interactions

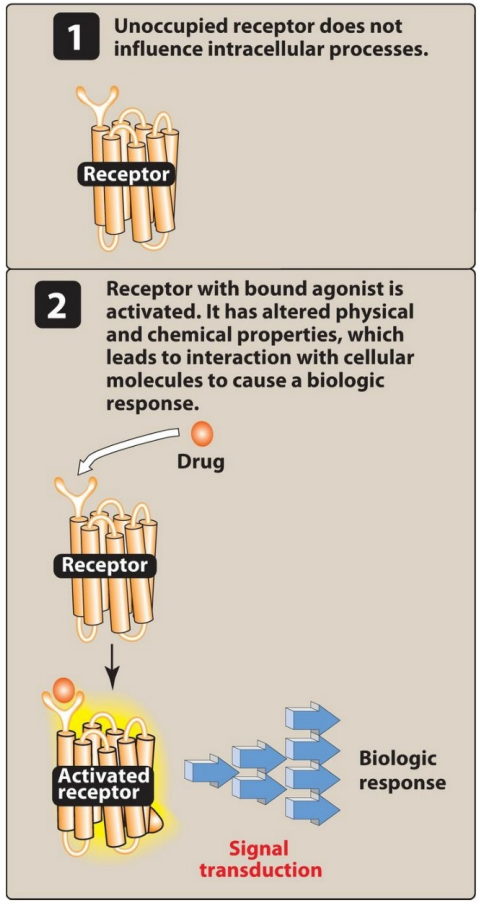
Hawler Medical University
College of Medicine
Department of Pharmacology
Dr. Susan Abdulkadir Farhadi
MSc, PhD Pharmacology



Objectives

- List different types of receptors
- Know the mechanism of drug receptor interaction
- Define certain pharmacodynamic terms

- Pharmacodynamics describes the actions of a drug on the body.
- Most drugs exert effects, both beneficial and harmful, by interacting with specialized target macromolecules called receptors, which are present on or in the cell.
- The drug–receptor complex initiates alterations in biochemical and/or molecular activity of a cell by a process called signal transduction



Signal Transduction

- Drugs act as signals, and receptors act as signal detectors.
- A drug is termed an "agonist" if it binds to a site on a receptor protein and activates it to initiate a series of reactions that ultimately result in a specific intracellular response.
- "Second messenger" or effector molecules are part of the cascade of events that translates agonist binding into a cellular response.

The drug–receptor complex

- Cells have many different types of receptors, each of which is specific for a particular agonist and produces a unique response.
- Cardiac cell membranes, for example, contain β-adrenergic receptors that bind and respond to epinephrine or norepinephrine.
- Cardiac cells also contain muscarinic receptors that bind and respond to acetylcholine.
- These two receptor populations dynamically interact to control the heart's vital functions.
- The magnitude of the cellular response is proportional to the number of drug–receptor complexes.
- This concept is conceptually similar to the formation of complexes between enzyme and substrate and shares many common features, such as specificity of the receptor for a given agonist.

Receptor states

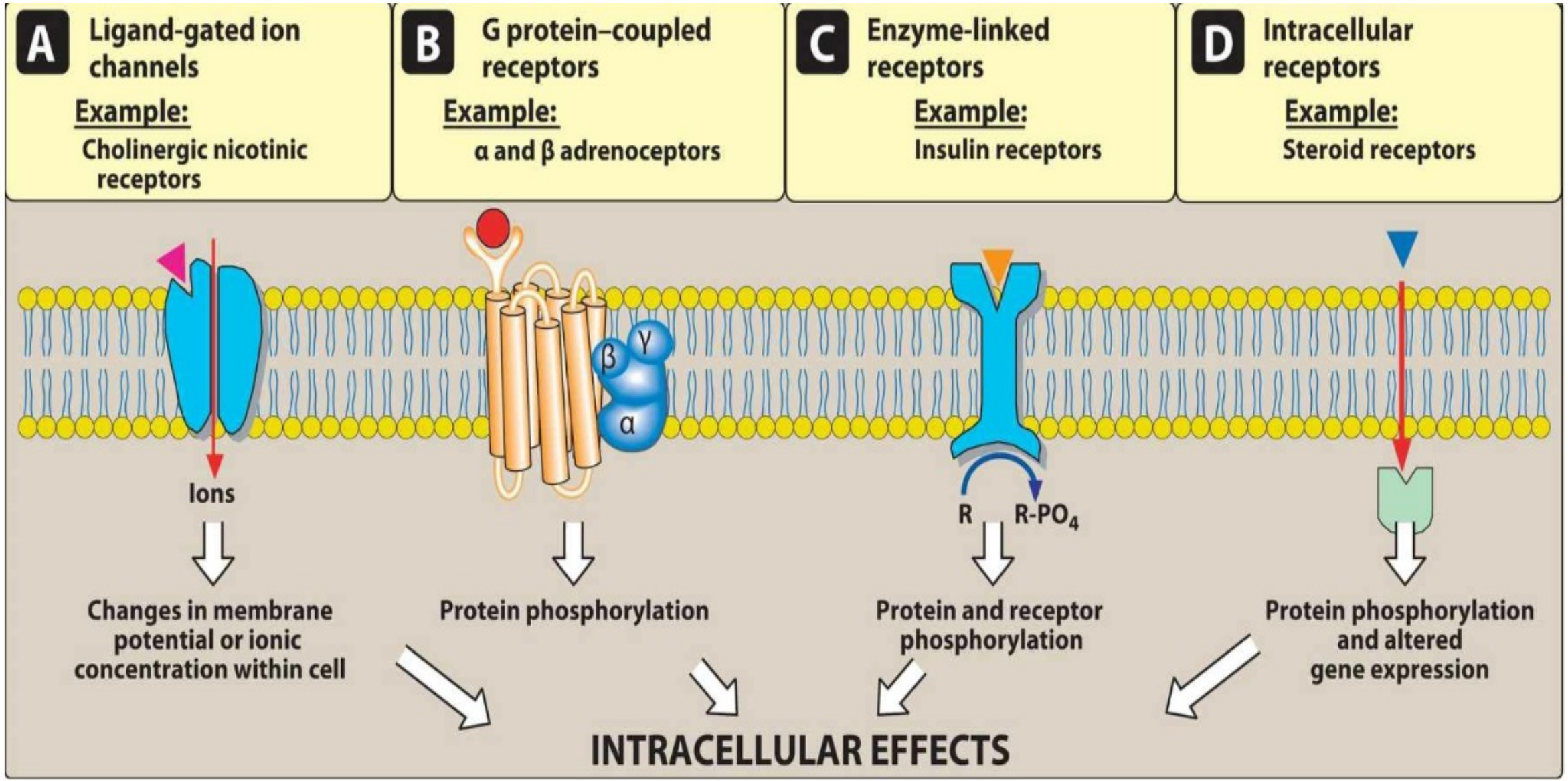
- Receptors exist in at least two states, inactive (R) and active (R*), that are in reversible equilibrium with one another, usually favoring the inactive state.
- Binding of agonists causes the equilibrium to shift from R to R* to produce a biologic effect.
- Antagonists are drugs that bind to the receptor but do not increase the fraction of R*, instead stabilizing the fraction of R.
- Some drugs (partial agonists) shift the equilibrium from R to R*, but the fraction of R* is less than that caused by an agonist.
- The magnitude of biological effect is directly related to the fraction of R*.
- So agonists, antagonists, and partial agonists are examples of molecules or ligands that bind to the activation site on the receptor and can affect the fraction of R*.

Major receptor families

- A receptor is defined as any biologic molecule to which a drug binds and produces a measurable response. Thus, enzymes, nucleic acids, and structural proteins can act as receptors for drugs or endogenous agonists.
- However, the richest sources of receptors are membrane-bound proteins that transduce extracellular signals into intracellular responses.
- These receptors may be divided into four families:

1) ligand-gated ion channels,
2) G protein–coupled receptors,
3) enzyme-linked receptors, and
4) intracellular receptors.

- Generally, hydrophilic ligands interact with receptors that are found on the cell surface.
- In contrast, hydrophobic ligands enter cells through the lipid bilayers of the cell membrane to interact with receptors found inside cells.


A
Ligand-gated ion channels
Example:
Cholinergic nicotinic receptors
B
G protein–coupled receptors
Example:
α and β adrenoceptors
C
Enzyme-linked receptors
Example:
Insulin receptors
D
Intracellular receptors
Example:
Steroid receptors
β
γ
α
Ions
R
R-PO4
Changes in membrane potential or ionic concentration within cell
Protein phosphorylation
Protein and receptor phosphorylation
Protein phosphorylation and altered gene expression
INTRACELLULAR EFFECTS

1. Transmembrane ligand-gated ion channels

- The extracellular portion of ligand-gated ion channels contains the drug-binding site.
- This site regulates the opening of the pore through which ions can flow across cell membranes .
- The channel is usually closed until the receptor is activated by an agonist, which opens the channel for a few milliseconds.
- Depending on the ion conducted through these channels, these receptors mediate diverse functions, including neurotransmission and muscle contraction. For example, stimulation of the nicotinic receptor by acetylcholine opens a channel that allows sodium influx and potassium outflux across the cell membranes of neurons or muscle cells.
- This change in ionic concentrations across the membrane generates an action potential in a neuron and contraction in skeletal and cardiac muscle.

- On the other hand, agonist stimulation of the A subtype of the γ-aminobutyric acid (GABA) receptor increases chloride influx, resulting in hyperpolarization of neurons and less chance of generating an action potential.
- Drug-binding sites are also found on many voltage-gated ion channels where they can regulate channel function.
- For example, local anesthetics bind to the voltage-gated sodium channel, inhibiting sodium influx and decreasing neuronal conduction.

2. Transmembrane G protein–coupled receptors

- The extracellular portion of this receptor contains the ligand-binding site, and the intracellular portion interacts (when activated) with a G protein.
- There are many kinds of G proteins (for example, Gs, Gi, and Gq), but all types are composed of three protein subunits.
- The α subunit binds guanosine triphosphate (GTP), and the β and γ subunits anchor the G protein in the cell membrane.

- Binding of an agonist to the receptor increases GTP binding to the α subunit, causing dissociation of the α-GTP complex from the βγ complex.
- The α and βγ subunits are then free to interact with specific cellular effectors, usually an enzyme or an ion channel, that cause further actions within the cell.
- These responses usually last several seconds to minutes. Often, the activated effectors produce “second messenger” molecules that further activate other effectors in the cell, causing a signal cascade effect.

- A common effector, activated by Gs and inhibited by Gi, is adenylyl cyclase, which produces the second messenger cyclic adenosine monophosphate (cAMP).
- The effector phospholipase C, when activated by Gq, generates two second messengers: inositol 1,4,5-trisphosphate (IP3) and diacylglycerol (DAG).
- DAG and cAMP activate specific protein kinases within the cell, leading to a myriad of physiological effects.
- IP3 increases intracellular calcium concentration, which in turn activates other protein kinases.

3. Enzyme-linked receptors

- This family of receptors undergoes conformational changes when activated by a ligand, resulting in increased intracellular enzyme activity.
- This response lasts for minutes to hours.
- The most common enzyme- linked receptors (for example, growth factors and insulin) possess tyrosine kinase activity.
- When activated, the receptor phosphorylates tyrosine residues on itself and other specific proteins.
- Phosphorylation can substantially modify the structure of the target protein, thereby acting as a molecular switch.
- For example, the phosphorylated insulin receptor in turn phosphorylates other proteins that now become active.
- Thus, enzyme-linked receptors often cause a signal cascade effect like that caused by G protein–coupled receptors.

4. Intracellular receptors

- The fourth family of receptors differs considerably from the other three in that the receptor is entirely intracellular, and, therefore, the ligand (for example, steroid hormones) must have sufficient lipid solubility to diffuse into the cell to interact with the receptor.
- The primary targets of activated intracellular receptors are transcription factors in the cell nucleus that regulate gene expression.

- The activation or inactivation of transcription factors alters the transcription of DNA into RNA and subsequently translation of RNA into proteins.
- The effect of drugs or endogenous ligands that activate intracellular receptors takes hours to days to occur.
- Other targets of intracellular ligands are structural proteins, enzymes, RNA, and ribosomes.

Characteristics of signal transduction

Signal transduction has two important features:

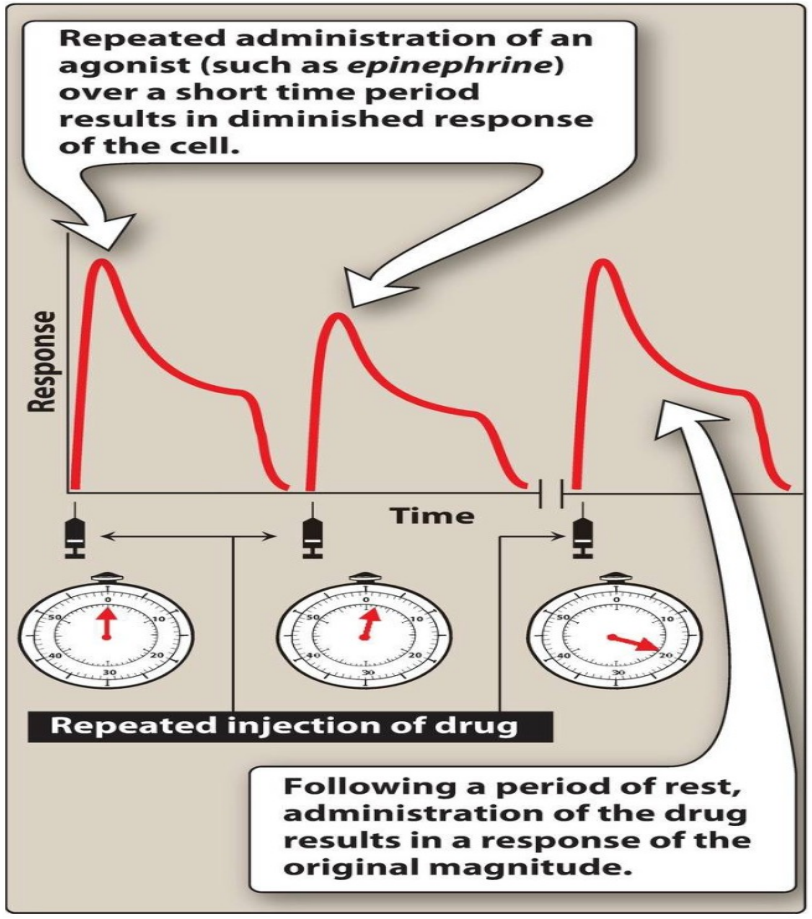
1) the ability to amplify small signals and
2) mechanisms to protect the cell from excessive stimulation.

1. Signal amplification

- A characteristic of G protein–linked and enzyme-linked receptors is the ability to amplify signal intensity and duration via the signal cascade effect.
- Additionally, activated G proteins persist for a longer duration than does the original agonist–receptor complex.
- The binding of albuterol, for example, may only exist for a few milliseconds, but the subsequent activated G proteins may last for hundreds of milliseconds.
- Further prolongation and amplification of the initial signal are mediated by the interaction between G proteins and their respective intracellular targets.
- Because of this amplification, only a fraction of the total receptors for a specific ligand may need to be occupied to elicit a maximal response.
- Systems that exhibit this behavior are said to have spare receptors. About 99% of insulin receptors are “spare,” providing an immense functional reserve that ensures that adequate amounts of glucose enter the cell.
- On the other hand, only about 5% to 10% of the total β-adrenoceptors in the heart are spare.
- Therefore, little functional reserve exists in the failing heart, because most receptors must be occupied to obtain maximum contractility.

2. Desensitization and down-regulation of receptors

- Repeated or continuous administration of an agonist or antagonist often leads to changes in the responsiveness of the receptor.
- The receptor may become desensitized due to too much agonist stimulation, resulting in a diminished response.
- This phenomenon, called tachyphylaxis, is often due to phosphorylation that renders receptors unresponsive to the agonist.
- In addition, receptors may be internalized within the cell, making them unavailable for further agonist interaction (down-regulation). Some receptors, particularly ion channels, require a finite time following stimulation before they can be activated again.
- During this recovery phase, unresponsive receptors are said to be "refractory."
- Repeated exposure of a receptor to an antagonist, on the other hand, results in up-regulation of receptors, in which receptor reserves are inserted into the membrane, increasing the number of receptors available.
- Up-regulation of receptors can make cells more sensitive to agonists and/or more resistant to effects of the antagonist.

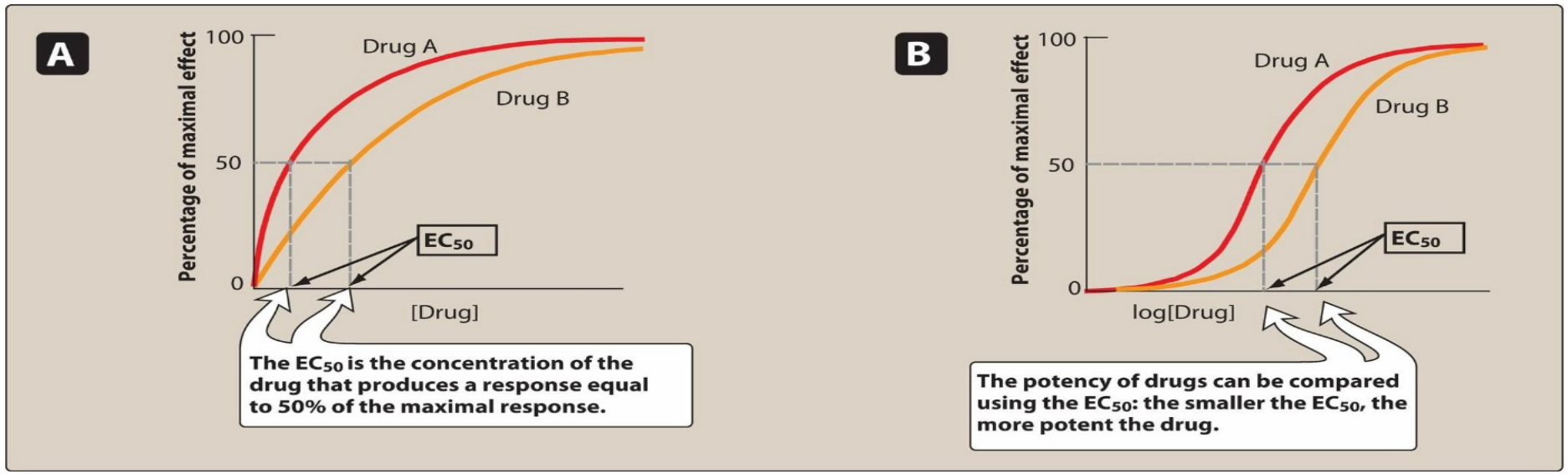


Dose–Response Relationships

- Agonist drugs mimic the action of the endogenous ligand for the receptor (for example, isoproterenol mimics norepinephrine on β1 receptors of the heart).
- The magnitude of the drug effect depends on receptor sensitivity to the drug and the drug concentration at the receptor site, which, in turn, is determined by both the dose of drug administered and by the drug's pharmacokinetic profile, such as rate of absorption, distribution, metabolism, and elimination.

Graded dose–response relationship

- As the concentration of a drug increases, its pharmacologic effect also gradually increases until all the receptors are occupied (the maximum effect).
- Plotting the magnitude of response against increasing doses of a drug produces a graded dose–response curve that has the general shape depicted in this diagram. Two important drug characteristics, potency and efficacy, can be determined by graded dose–response curves.

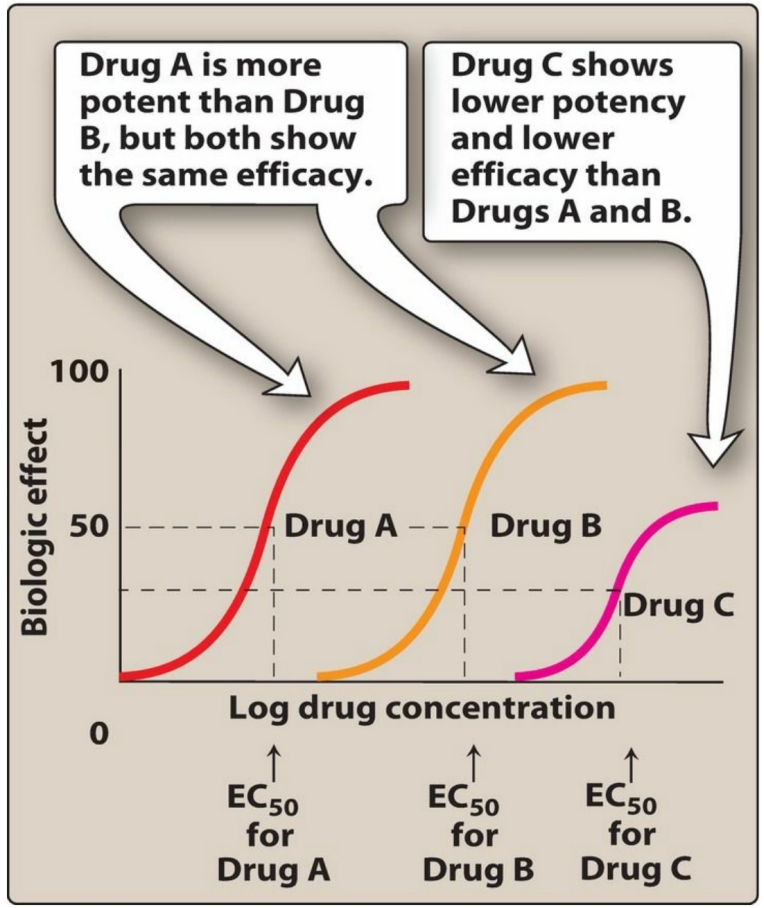


1. Potency

- Potency is a measure of the amount of drug necessary to produce an effect.
- The concentration of drug producing 50% of the maximum effect (EC50) is often used to determine potency.
- In the previous figure, the EC50 for Drugs A and B indicate that Drug A is more potent than Drug B, because a lesser amount of Drug A is needed to obtain 50% effect.
- Therapeutic preparations of drugs reflect their potency. For example, candesartan and irbesartan are angiotensin receptor blockers used to treat hypertension. The therapeutic dose range for candesartan is 4 to 32 mg, as compared to 75 to 300 mg for irbesartan. Therefore, candesartan is more potent than irbesartan (it has a lower EC50 value).

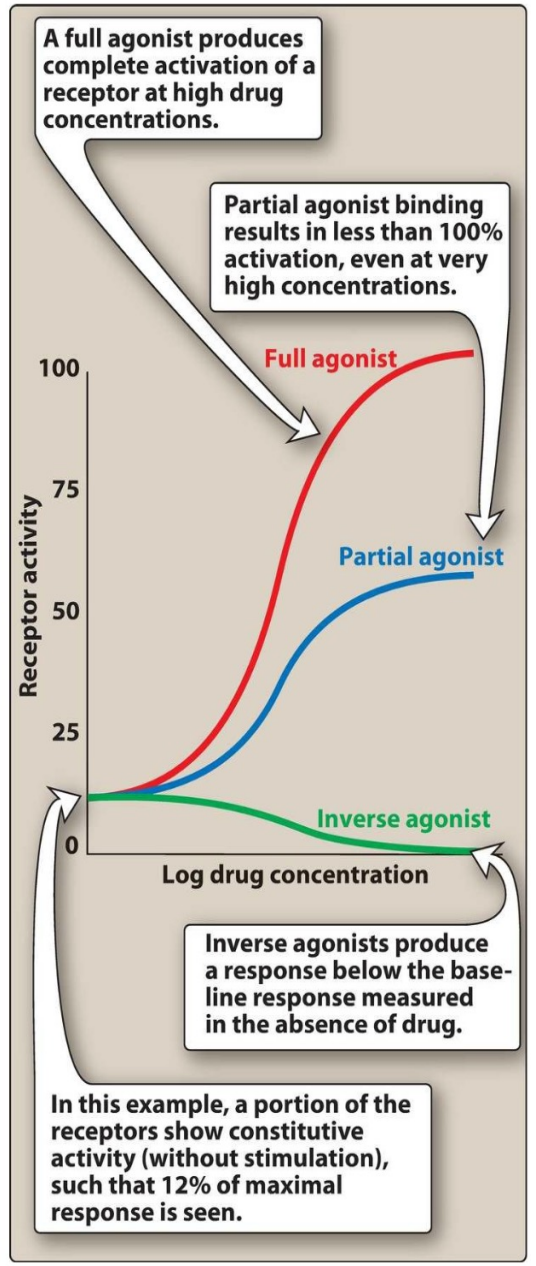
2. Efficacy

- Efficacy is the magnitude of response a drug causes when it interacts with a receptor.
- Efficacy is dependent on the number of drug–receptor complexes formed and the intrinsic activity of the drug (its ability to activate the receptor and cause a cellular response).
- Maximal efficacy of a drug (Emax) assumes that the drug occupies all receptors, and no increase in response is observed in response to higher concentrations of drug.
- The maximal response differs between full and partial agonists, even when the drug occupies 100% of the receptors.
- Similarly, even though an antagonist occupies 100% of the receptor sites, no receptor activation results and Emax is zero.
- Efficacy is a more clinically useful characteristic than potency, since a drug with greater efficacy is more therapeutically beneficial than one that is more potent



A. Full agonists

- If a drug binds to a receptor and produces a maximal biologic response that mimics the response to the endogenous ligand, it is a full agonist.
- Full agonists bind to a receptor, stabilizing the receptor in its active state and are said to have an intrinsic activity of one.
- All full agonists for a receptor population should produce the same Emax.
- For example, phenylephrine is a full agonist at α1-adrenoceptors, because it produces the same Emax as the endogenous ligand, norepinephrine.
- Upon binding to α1-adrenoceptors on vascular smooth muscle, both norepinephrine and phenylephrine stabilize the receptor in its active state, thereby increasing Gq activation.
- Activation of Gq increases intracellular Ca2+, causing interaction of actin and myosin filaments and shortening of the muscle cells.
- The diameter of the arteriole decreases, causing an increase in resistance to blood flow through the vessel and an increase in blood pressure.
- Thus, effects of agonists on intracellular molecules, cells, tissues, and intact organisms are all attributable to interaction of the drug with the receptor.



B. Partial agonists

- Partial agonists have intrinsic activities greater than zero but less than one .
- Even when all the receptors are occupied, partial agonists cannot produce the same Emax as a full agonist.
- Even so, a partial agonist may have an affinity that is greater than, less than, or equivalent to that of a full agonist.
- A partial agonist may also act as a partial antagonist of a full agonist .
- As the number of receptors occupied by the partial agonist increases, the number of receptors that can be occupied by the full agonist decreases and therefore Emax would decrease until it reached the Emax of the partial agonist.
- This potential of partial agonists to act as both an agonist and antagonist may have therapeutic utility.

C. Inverse agonists

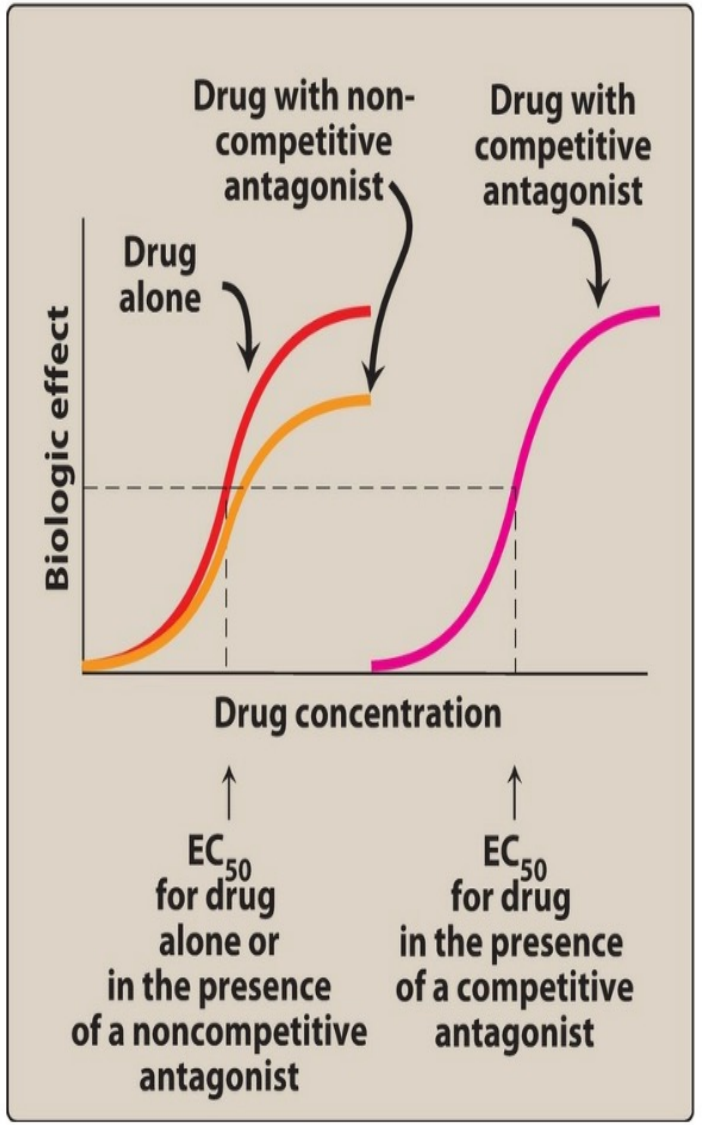
- Typically, unbound receptors are inactive and require interaction with an agonist to assume an active conformation.
- However, some receptors show a spontaneous conversion from R to R* in the absence of an agonist.
- Inverse agonists, unlike full agonists, stabilize the inactive R form and cause R* to convert to R.
- This decreases the number of activated receptors to below that observed in the absence of drug.
- Thus, inverse agonists have an intrinsic activity less than zero, reverse the activation state of receptors, and exert the opposite pharmacological effect of agonists.

D. Antagonists

- Antagonists bind to a receptor with high affinity but possess zero intrinsic activity.
- An antagonist has no effect on biological function in the absence of an agonist, but can decrease the effect of an agonist when present.
- Antagonism may occur either by blocking the drug's ability to bind to the receptor or by blocking its ability to activate the receptor.

1. Competitive antagonists

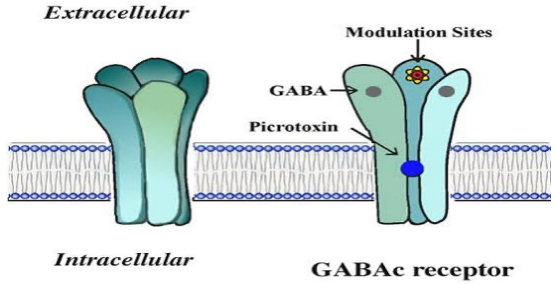
- If the antagonist binds to the same site on the receptor as the agonist in a reversible manner, it is “competitive.”
- A competitive antagonist interferes with an agonist binding to its receptor and maintains the receptor in its inactive state.
- For example, the antihypertensive drug terazosin competes with the endogenous ligand norepinephrine at α1-adrenoceptors, thus decreasing vascular smooth muscle tone and reducing blood pressure.
- However, increasing the concentration of agonist relative to antagonist can overcome this inhibition.
- Thus, competitive antagonists characteristically shift the agonist dose–response curve to the right (increased EC50) without affecting Emax.



2. Irreversible antagonists

- Irreversible antagonists bind covalently to the active site of the receptor, thereby permanently reducing the number of receptors available to the agonist.
- An irreversible antagonist causes a downward shift of the Emax, with no shift of EC50 values.
- In contrast to competitive antagonists, addition of more agonist does not overcome the effect of irreversible antagonists.
- Thus, irreversible antagonists and allosteric antagonists are both considered noncompetitive antagonists.
- A fundamental difference between competitive and noncompetitive antagonists is that competitive antagonists reduce agonist potency (increase EC50) and noncompetitive antagonists reduce agonist efficacy (decrease Emax).

3. Allosteric antagonists



- An allosteric antagonist binds to a site (allosteric site) other than the agonist-binding site and prevents receptor activation by the agonist.
- This type of antagonist also causes a downward shift of the Emax of an agonist, with no change in the EC50 value.
- An example of an allosteric agonist is picrotoxin, which binds to the inside of the GABA-controlled chloride channel. When picrotoxin binds inside the channel, no chloride can pass through the channel, even when GABA fully occupies the receptor.

4. Functional antagonism

- An antagonist may act at a completely separate receptor, initiating effects that are functionally opposite those of the agonist.
- A classic example is the functional antagonism by epinephrine to histamine-induced bronchoconstriction.
- Histamine binds to H1 histamine receptors on bronchial smooth muscle, causing bronchoconstriction of the bronchial tree. Epinephrine is an agonist at β2-adrenoceptors on bronchial smooth muscle, which causes the muscles to relax.
- This functional antagonism is also known as “physiologic antagonism”.

References

- Lippincott Illustrated Review - Pharmacology 7th Edition (2018)-3